Full Stack Development with MERN

Project Documentation

1. Introduction

Project Title: Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Team Members:

- Jannath Anisha S – Project Developer
- **Team Members:** Not Specified

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2. Project Overview

Purpose:

The purpose of this project is to import structured employee data from an Excel spreadsheet into ServiceNow using **Import Sets and Transform Maps**. The project helps automate employee data entry, maintain accurate records, prevent duplicate records, and validate imported data.

Features:

- Upload employee data from an Excel spreadsheet.
- Create and use an Import Set table.
- Create a Transform Map for employee data.
- Map spreadsheet fields to target table fields.
- Transform employee data into the Employee Test table.
- Use Coalesce to prevent duplicate records.
- Update existing employee records.
- Verify import results using Transform History.
- Generate employee reports.
- Create an Employee Analytics Dashboard.

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3. Architecture

Frontend:

The project uses the **ServiceNow Web User Interface** through which the HR/Admin user can upload employee spreadsheet data, configure Import Sets and Transform Maps, view employee records, generate reports, and access the dashboard.

Backend:

The application logic is handled by the **ServiceNow platform**. Import Sets are used to load spreadsheet data into a staging table, while Transform Maps are used to transfer and transform the data into the target Employee Test table.

Database:

The project uses the **ServiceNow database** to store employee information. The main tables used are:

- **Employee Import** – Staging table for imported spreadsheet data.
- **Employee Test** – Target table containing transformed employee records.

The Employee Test table contains fields such as:

- Employee ID
- Employee Name
- Email
- Department
- Location

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4. Setup Instructions

Prerequisites:

- ServiceNow instance
- Web browser
- Employee Excel spreadsheet
- Authorized ServiceNow user account

Installation / Setup:

1. Open the ServiceNow instance.
2. Log in using authorized credentials.
3. Prepare the employee data in an Excel spreadsheet.
4. Create the **Employee Test** target table.
5. Create the **Employee Import** Import Set table.
6. Upload the employee spreadsheet using **Load Data**.
7. Create the **Sample Spreadsheet Import** Transform Map.
8. Select **Employee Import** as the Source Table.
9. Select **Employee Test** as the Target Table.
10. Map the employee fields.
11. Enable Coalesce for the required field.
12. Run the transformation.
13. Verify the imported records.
14. Create reports and the Employee Analytics Dashboard.

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5. Folder Structure

This project is developed within the **ServiceNow platform**, so a traditional React client/server folder structure is not used.

Client:

The ServiceNow Web UI provides the user interface for:

- Uploading spreadsheets
- Managing Import Sets
- Configuring Transform Maps

- Viewing employee records
- Viewing reports and dashboards

Server:

The ServiceNow platform handles:

- Import Set processing
- Transform Map execution
- Field mapping
- Coalesce processing
- Employee record creation and updates
- Transform History
- Reports and dashboard data

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6. Running the Application

The project runs through the **ServiceNow instance**.

Steps:

1. Open a web browser.
2. Open the ServiceNow instance.
3. Log in with authorized credentials.
4. Navigate to the required Import Set and Transform Map modules.
5. Upload the employee spreadsheet.
6. Run the transformation.
7. Open the Employee Test table to verify the records.
8. Open Transform History to check the import results.
9. Open the Employee Analytics Dashboard to view the reports.

Note: npm start commands are not applicable because this project is implemented using ServiceNow Import Sets and Transform Maps rather than a MERN application.

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7. API Documentation

This project does not use custom REST APIs or external APIs.

The employee data processing is performed using built-in ServiceNow functionality:

ServiceNow Feature	Purpose
Import Set	Loads employee spreadsheet data into a staging table.
Transform Map	Transfers and transforms data from the source table to the target table.
Field Mapping	Maps source spreadsheet fields to target employee fields.
Coalesce	Identifies existing records and prevents unnecessary duplicate records.
Transform History	Displays the results of data transformation.
Reports	Provides employee data analysis.
Dashboard	Displays employee reports in one place.

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8. Authentication

Access to the project is provided through the **ServiceNow user authentication system**.

Authorized users log in to the ServiceNow instance using their credentials. Access to employee data and project features is controlled through ServiceNow user roles and access controls.

The project does not implement separate JWT tokens, sessions, or custom authentication because authentication is handled by the ServiceNow platform.

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9. User Interface

The following ServiceNow screens can be included as screenshots in the documentation:

1. **Employee Excel Spreadsheet**
2. **Employee Test Table**
3. **Employee Import / Load Data**
4. **Sample Spreadsheet Import Transform Map**
5. **Field Mapping / Mapping Assist**
6. **Transform Success Result**
7. **Coalesce Configuration**
8. **Transform History**
9. **Employee Reports**
10. **Employee Analytics Dashboard**

These screenshots demonstrate the complete employee data import and transformation process.

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10. Testing

The project was tested using User Acceptance Testing (UAT).

Testing Areas:

- Employee spreadsheet upload
- Import Set data loading
- Transform Map configuration
- Field mapping
- Data transformation

- New employee record insertion
- Existing employee record update
- Duplicate record prevention using Coalesce
- Transform History validation
- Department report
- Location report
- Employee List Report
- Employee Analytics Dashboard

Testing Result:

The major functional test cases were successfully completed. Employee data was imported, transformed, validated, and displayed through reports and the dashboard.

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12. Known Issues

- The project depends on correctly formatted employee spreadsheet data.
- Incorrect or missing spreadsheet fields may affect the transformation.
- Incorrect field mapping can result in incorrect target records.
- The ServiceNow instance must be accessible for performing imports and viewing reports.
- Unauthorized users may not be able to access employee data or project features.

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13. Future Enhancements

The following enhancements can be considered in future versions:

- Add automated scheduled employee data imports.
- Add email notifications after successful or failed imports.
- Add additional employee data validation rules.
- Add more employee analytics and reports.

- Add advanced dashboard filters.
- Add automated error reporting for invalid spreadsheet records.
- Support additional spreadsheet formats.
- Add more detailed import monitoring and audit information.